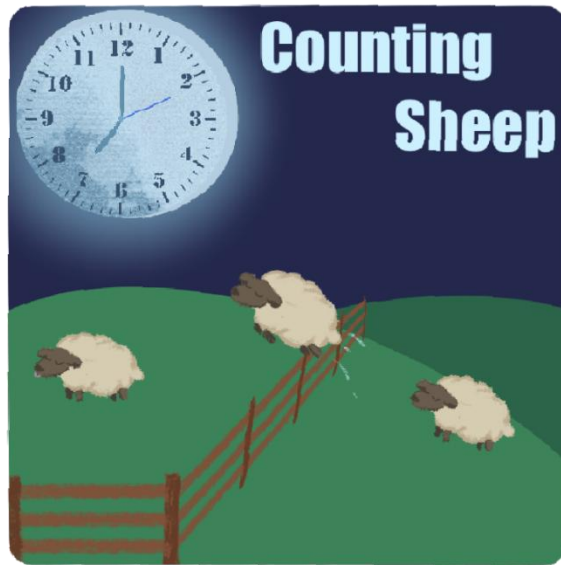


Counting Sheep
Software Requirement Specification

Fall 2025

Version 1.0



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1. Introduction

1.1. Purpose

This Software Requirement Specification is designed to outline the requirements and goals of the Counting Sheep Application. Requirements listed in this document reflect those discussed with the client Dr. Jin. These requirements are expected to be implemented and tested by the end of the Fall 2025 Semester for the CSE-4550 Software Engineering Course

1.2. Scope

The Counting Sheep Application will enable users to not only monitor and track their sleep habits but will also provide the user with a gamified way of improving their current sleep habits. The main elements of this are the Android Application and database to access other information.

Prototype 1: The first prototype will consist of an application which will allow users to login in or create a new user profile, a functionable general user homepage, functionable assessment interface, functional exercises interface.

Prototype 2: The second prototype will consist of the application which will allow users to view their sleep goal history, an admin interface, a functionable add/remove client interface, and a functional view user goal interface.

1.3 Definitions, Acronyms, and Abbreviations

Android: Mobile operating system running on the user's device.

App: A type of software that allows the user to perform specific tasks.

C#: A general-purpose high-level programming language

Mobile device: A portable computing device such as a smartphone or tablet computer.

Deployment diagram: A structure diagram which shows the architecture of the system as deployment (distribution) of software artifacts to deployment targets.

Emoji: A small digital image or icon used to express an idea, emotion, etc., in electronic communication.

Firebase Web Server: Web based server provided by google which allows access to web services and database

Firebase Database: Location where all user data is stored and accessed during use of the application

Home screen: The introductory interface displayed on the device, from which a user can access functions.

IDE: Integrated Development Environment, used by programmers to test and develop code in a specific language for a device.

Programming Language: A vocabulary and set of grammatical rules for instructing a computer or computing device to perform specific tasks.

SRS: This document, the software requirements specification (SRS) document lays out the functional and non-functional requirements of the Counting Sheep App.

UML: Unified Modeling Language (UML) is a standardized modeling language enabling developers to specify, visualize, construct and document artifacts of a software system.

Unity: A cross-platform game engine particularly popular for developing 2D and 3D games on platforms such as Android

Use case diagram: A representation of a user's interaction with the system that shows the relationship between the user and the different use cases in which the user is involved.

User: Someone who uses the application or software specified in this software requirements specification.

1.4 References

[1] IEEE Software Engineering Standard Committee, "IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications", October 20, 1998.

[2] Fowler, Martin. *UML Distilled: A brief guide to the standard object modeling language - third edition*. Pearson Education, Inc. 25 September 2003.

[3] "Unity Real-Time Development Platform: 3D, 2D, VR & AR Engine." *Unity*, unity.com/. Accessed 28 Sept. 2025.

[4] "Firebase Cloud Database and Web Server." *Google*, Google, firebase.google.com/. Accessed 28 Sept. 2025.

[5] "Diagramming Powered by Intelligence." *Lucidchart*, www.lucidchart.com/pages. Accessed 28 Sept. 2025.

[6] "GitHub · Build and Ship Software on a Single, Collaborative Platform." *GitHub*, github.com/. Accessed 28 Sept. 2025.

1.5 Overview

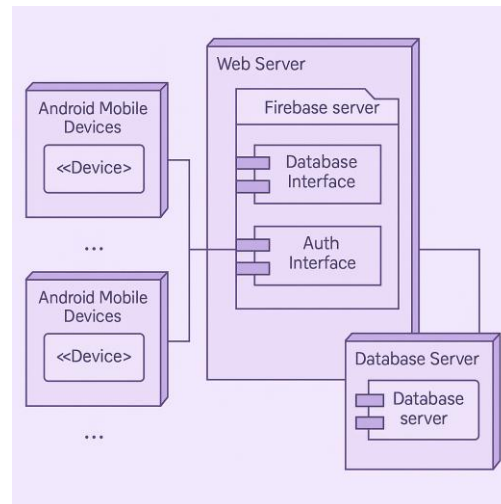
The following information provided in this document is intended to provide further details about the application as well as provide details of the usage, functionality and plans for the Counting Sheep application.

The second section of this document provides the user with an overview of the application and how it works. It describes how the software systems are integrated and their constraints.

The third section of this document specifies the requirements of the project. It demonstrates specific components, iterations of software interfaces, design constraints, and system requirements.

2. Overall Description

2.1.1 Product Perspectives System interfaces (deployment diagram)



The major components involved in this project include the Firebase web server which contains interfaces for management, the Firebase database which provides real-time database and the android device itself.

2.1.2 User interfaces

Individual User

Upon opening the Counting Sheep application, the user will be taken to a user login screen. If the user is already logged in, they will automatically be taken to the user home screen. The home screen displays the company's logo with a variety of options for the user including, set goals, view goals, log sleep and settings. The assessment screen displays a version of the company logo with four clickable regions. Once the user finishes their goals they will be taken to the view sleep interface where they will be given information to improve their sleep habits. The view history option will allow users to track and monitor their sleep goals over a period of time. The settings interface will allow users to change their goals, notifications, or to log out.

Software interfaces

There are two major software interfaces used in this project

1. **Unity** is used to work on mobile applications for Android Devices
2. **Firebase Server** is used to handle HTTP requests sent by Android Devices.

The Firebase Server runs on an Occupancy Web Server which includes two interfaces

1. **Authentication** - which stores user emails and assigns them a UID
2. **Database** - web based database which stores and organizes user information.

2.1.3 Communication interfaces

The only communication interface is between the mobile device and the Firebase Server. The mobile device establishes an HTTP connection with the Firebase Web Server which can be relayed back to the mobile device.

2.1.4 Memory

The Firebase Server is limited to 1GB of storage and 10 GB downloads with 100 simultaneous connections. This is due to using the free services that Firebase provides. If the app is completed clients may view service plans for larger storage capacity. The mobile app should be limited to less than 5GB of memory.

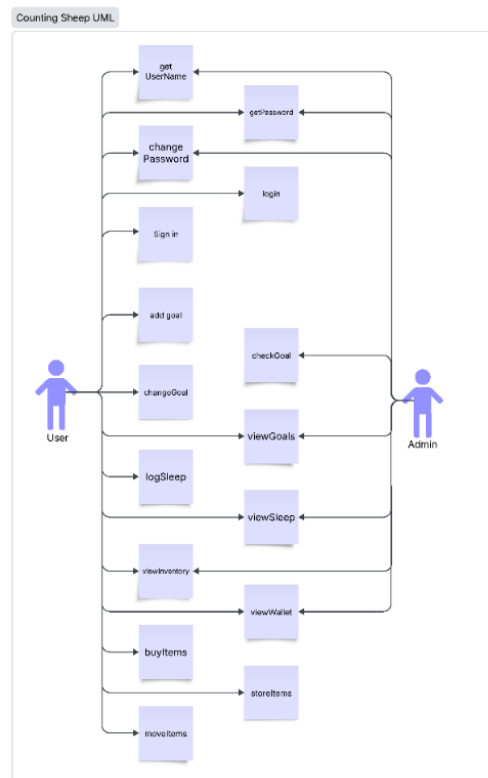
2.1.5 Operation

The Firebase Web Server will need to be active during all activity of the mobile app. As Firebase is provided by google Inc., their servers run 24/7 and provide adequate protection of data.

2.2 Product Functions (use case diagram)

The application allows users to track their sleep goals, while attempting to provide solutions for them to improve their current sleep habits. The admin side of the application allows admins to view the current sleep habits of their users which will help them in providing them with information which could help aid their users.

(Figure 2.3)



2.3 User Characteristics

The Use-Case Diagram shows that the personal user will be able to take assessments, view exercises to improve their mental state, review their past goals, and change personal settings. The admin user will be able to add/remove users, view their user's goals, and edit personal settings.

The intended demographic using the Counting Sheep application encompasses all individuals who would like to monitor or improve their sleep habits. The application does, however, allow admins to see users' goals, progress, and wallet balance.

2.4 Constraints

1. Learning unfamiliar technologies such as Server, databases, and Unity.
2. Becoming familiar with Firebase and all its features
3. Setting up a Server and database in a timely manner.
4. Learning how to properly design and develop an application, in a professional setting
5. Managing busy schedules and other responsibilities

2.5 Assumptions and Dependencies

1. The Android Device has internet access
2. Firebase features such as database and server will always be functional

3. Specific Requirements

3.1 External interface requirements

3.1.1 User interfaces

The user interface consists of graphical elements in the Android App. More information of the app's interface is found in Section 3.2

3.1.2 Hardware interfaces

There are no external hardware interfaces for this project

3.1.3 Software interfaces

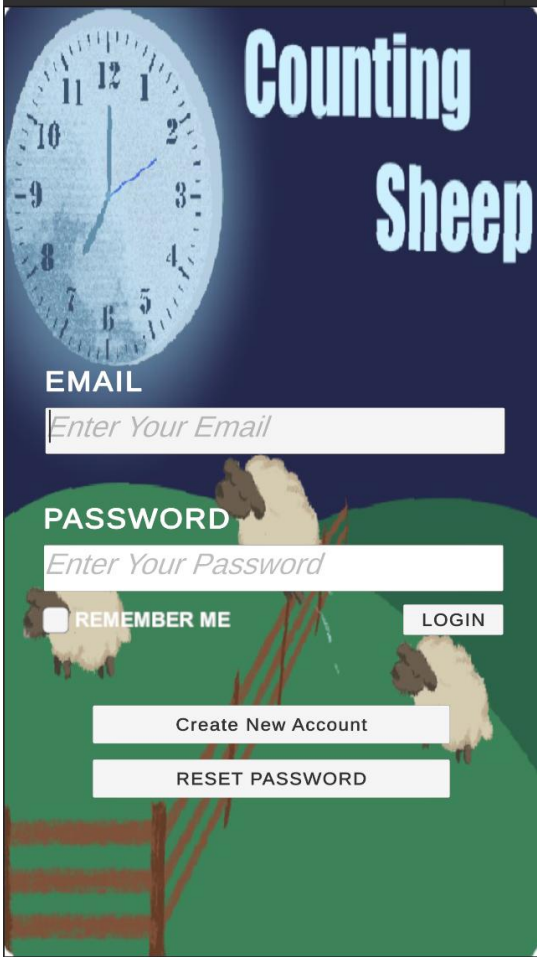
The Counting Sheep app communicates with the Firebase Web Server to retrieve and save information for the user, using Firebase Database.

Communication interfaces The Firebase Web Server must communicate with mobile devices to send and retrieve user's information on the Android device.

3.2 Functional requirements and User Interfaces

If a user is not already logged into the application, the login interface will be first to load (Figure1)

(Figure 1: Login Screen)



The login screen features a dark blue background with a large, glowing clock face in the upper left corner. The title "Counting Sheep" is displayed in a large, light blue, stylized font in the upper right. Below the clock, the word "EMAIL" is written in white, followed by a text input field with the placeholder text "Enter Your Email". Below this, the word "PASSWORD" is written in white, followed by a text input field with the placeholder text "Enter Your Password". To the left of the password field is a checkbox labeled "REMEMBER ME". To the right of the password field is a button labeled "LOGIN". Below these fields are two more buttons: "Create New Account" and "RESET PASSWORD". The background also features a green field with a brown fence and two sheep.

Counting Sheep

EMAIL

Enter Your Email

PASSWORD

Enter Your Password

☐ REMEMBER ME

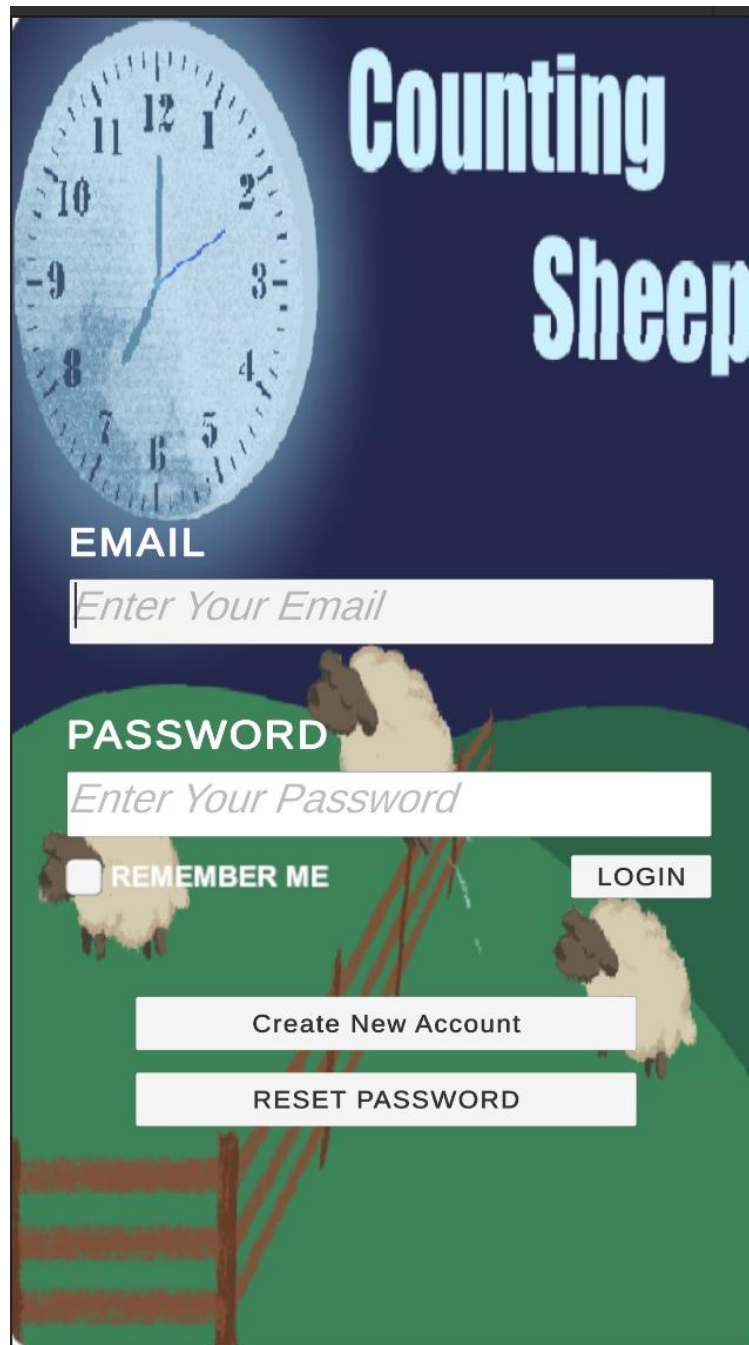
LOGIN

Create New Account

RESET PASSWORD

If a user does not have an account, they may click the create account button which will take them to the create account interface. (Figure 2)

(Figure 2: Create Account Screen)



The image shows a 'Create Account' screen for an application titled 'Counting Sheep'. The background is a dark blue night sky with a large, glowing clock face in the upper left. The clock face has numbers 1 through 12, with the hands pointing to approximately 10:10. The title 'Counting Sheep' is written in a large, light blue, stylized font in the upper right. Below the clock, the word 'EMAIL' is displayed in white, followed by a white input field containing the placeholder text 'Enter Your Email'. Below this, the word 'PASSWORD' is displayed in white, followed by a white input field containing the placeholder text 'Enter Your Password'. To the left of the password field is a checkbox labeled 'REMEMBER ME'. To the right of the password field is a white button labeled 'LOGIN'. Below the 'REMEMBER ME' checkbox and the 'LOGIN' button are two more white buttons: 'Create New Account' and 'RESET PASSWORD'. The bottom of the screen features a green field with a brown fence and two sheep, one on the left and one on the right.

Counting Sheep

EMAIL

Enter Your Email

PASSWORD

Enter Your Password

☐ REMEMBER ME

LOGIN

Create New Account

RESET PASSWORD

3.3 Performance requirements

The Firebase Web server must update user information in real time to provide admins with up-to-date information, as well as to update recommended goals provided to general users. Depending on their updated status or on updated data provided content recommendations to users will change, thus real time updates are important.

3.4 Design constraints

The Mobile app must be connected to the internet to be able access the database and retrieve user information. Without internet access the application will not be able to retrieve data required to function properly.

3.5 Software system attributes

3.5.1 Reliability

Measures will be taken to ensure the reliability of the application and that user data is secure. This means ensuring that only those who are authorized to view users information will be able to do so.

3.5.2 Maintainability

Software will be sufficiently documented such that the application will be easy to maintain and modify. Future features which could be implemented will be kept in mind, thus development will be very detailed and allow for future modification which clients have expressed they might like to include.

3.5.3 Portability

Current Android app cannot be ported to another language as this time.

Signature: _____

Date: _____

